

Public Signals, Reflexive Adoption, and Forecast Erosion in a Single-Asset Market

A Focused Research Proposal in Econometrics, Agent-Based Computational Finance, and Predictive Modeling

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v2 incorporates the revised §3.1 economic microfoundation (Brock–Hommes demand, Beja–Goldman / Farmer–Joshi market maker) and a tightened metric set.

Abstract

This proposal studies a simple but consequential question for modern financial markets: can a forecasting rule lose predictive and economic value *because* enough market participants adopt it and act on it? The project develops a disciplined single-asset agent-based market in which traders may switch from a null trading rule to a shared autoregressive forecasting rule. Individual demand is microfounded as mean-variance with constant absolute risk aversion (Brock and Hommes, 1998), and price formation follows the disequilibrium market-maker tradition of Beja and Goldman (1980) formalised in discrete log-linear form by Farmer and Joshi (2002). The central mechanism is *reflexive adoption*: as a predictive signal diffuses across traders, their orders become more coordinated, contemporaneous price impact absorbs more of the predictable movement, and the next-period predictable component available to the rule shrinks. The contribution is deliberately narrow. Rather than attempting a whole-economy or high-dimensional machine-learning programme, the proposal isolates one tractable mechanism in a transparent simulation environment, defines clear benchmarks, and uses a staged implementation plan that is feasible for a Master's capstone. The framework is intended to distinguish forecast erosion that arises endogenously from deployment from ordinary forecast instability.

Core research question. Under what combinations of adoption share, price impact, and retraining frequency does a shared autoregressive forecasting rule lose out-of-sample predictive power and net trading value *because* its own use crowds the signal out?

1. Introduction

Predictive models are now deeply embedded in financial markets. They appear in discretionary investment research, quantitative trading, portfolio construction, and automated decision systems. Most evaluation frameworks treat forecasting as a problem of learning a stable relation from historical data and then applying it to new observations. That approach is often useful, but it becomes fragile when the use of a forecast changes the market environment on which the forecast is evaluated.

The proposal addresses that problem in a focused setting: a discrete-time market for one risky asset, populated by interacting traders. The core intuition is straightforward. Suppose a forecasting rule identifies a weak but usable pattern in returns. When only a few traders use the rule, the pattern may retain predictive and trading value. When many traders use the same rule, however, their trades become more similar. That synchronization can alter order flow, move prices sooner, change volatility, and weaken the very signal that initially appeared profitable. In that sense, predictive success may contribute to its own erosion.

This question matters for two reasons. First, it speaks directly to the reliability of predictive systems in markets where information diffuses quickly and strategies are often imitated. Second, it links several important literatures that are rarely integrated operationally: structural instability in econometrics, heterogeneous-agent finance, performative prediction in machine learning, and empirical evidence on signal crowding and post-publication decay (Lucas 1976; Brock and Hommes 1998; Farmer and Joshi 2002; McLean and Pontiff 2016; Perdomo et al. 2020; Khandani and Lo 2011). The proposal preserves that intellectual ambition but places it inside a narrow and feasible design.

The scope is intentionally disciplined. The project does *not* attempt to model the whole economy, estimate a realistic market microstructure model, or compare a large menu of forecasting architectures. Instead, it focuses on one transparent mechanism in one market: a shared forecasting rule can become economically consequential enough to change the return process it exploits. The expected contribution is a minimal but rigorous simulation framework that can distinguish ordinary forecast instability from deployment-induced erosion, quantify when that erosion emerges, and identify which ingredients appear necessary for it.

2. Background and motivation

2.1 Key concepts

An *agent-based model* is a simulation in which the aggregate behaviour of a market is generated from the interactions of individual decision-makers, called agents. This approach is especially useful when heterogeneity, feedback, and strategic interaction are central to the question rather than secondary complications (Farmer and Foley 2009; Axtell and Farmer 2025). In the present context, an agent-based model is attractive because forecast erosion is not a property of one trader alone; it is a property of many traders responding to a common signal.

Heterogeneous agents are agents who do not all use the same rule. In finance, this matters because market outcomes depend on how different beliefs and strategies interact. A market with only one type of trader cannot represent imitation, crowding, or switching between rules in a meaningful way. The proposal uses heterogeneity parsimoniously: at baseline, traders differ only in whether they have adopted the shared forecasting rule; richer trader ecologies are reintroduced as extensions.

Forecasting in economics and finance means predicting a future variable using currently available information. Here the target is the next-period return. If p_t denotes the log-price of

the asset at time t , the one-period return is defined as $r_{t+1} = p_{t+1} - p_t$. Forecasting is difficult because useful predictive relations in financial data are often weak, unstable, and sensitive to behavioural change (Rossi 2021).

An *autoregressive model* uses past values of a variable to predict its next value. For example, an AR(1) model predicts r_{t+1} using r_t . Such models are a natural starting point for this project because they are transparent, easy to estimate with rolling windows, and directly interpretable in econometric terms. They also provide a clean benchmark before considering richer machine-learning extensions (Mullainathan and Spiess 2017).

Reflexivity refers to situations in which beliefs and actions help shape the environment they are meant to describe. In financial markets this means that traders do not merely observe prices; through their orders they also contribute to price formation. The current proposal uses the term in a precise and limited way: a forecast influences trading, trading influences returns, and returns then determine whether the forecast continues to perform well (Lo 2004; MacKenzie 2006).

Performative prediction is the machine-learning term for prediction problems in which a deployed model changes the future data on which it is evaluated. The core conceptual contribution of that literature is that accuracy can no longer be assessed as though the model were a passive observer (Perdomo et al. 2020; Hardt and Mendler-Dünner 2025). A public trading signal is performative if acting on it changes subsequent returns.

Model crowding occurs when many market participants rely on the same rule, signal, or model. In practice, crowding can manifest as similar positions, higher turnover in the same direction, and lower profitability of once-useful signals. In the proposed framework, crowding is measured through adoption rates, correlation of positions, and changes in forecast performance (Khandani and Lo 2011).

Endogenous concept drift describes a change in the relationship between predictors and outcomes that arises from the system itself rather than from an external shock. In this proposal, the relation between lagged returns and future returns may drift because traders adopt the forecasting rule and thereby alter the market process that generated the original predictability.

2.2 Relation to existing literature

The proposal draws on four strands of literature.

First, the econometric literature has long emphasised that historically estimated relations need not remain valid after behaviour changes. The classic statement is the Lucas critique (Lucas 1976). More recent work shows that forecast breakdowns and predictive instabilities are common in macro-financial settings (Giacomini and Rossi 2009; Rossi 2021). This literature motivates the need to separate ordinary instability from deterioration created by deployment itself.

Second, heterogeneous-agent finance demonstrates that interacting traders with different rules can generate nonlinear dynamics, instability, and regime change. The seminal framework of Brock and Hommes (1998) is particularly relevant here because it links market outcomes to the relative performance of competing forecasting rules, and it supplies the

mean-variance microfoundation used in §3.1. Price formation is taken from the disequilibrium market-maker tradition of Beja and Goldman (1980), formalised in discrete log-linear form by Farmer and Joshi (2002). More broadly, agent-based modelling has matured into a serious methodology for economics and finance when feedback and interaction are central (Farmer and Foley 2009; Axtell and Farmer 2025; Grazzini et al. 2017).

Third, performative prediction in machine learning offers a formal language for systems in which deployed models alter future data distributions (Perdomo et al. 2020; Hardt and Mendler-Dünnér 2025). That literature does not by itself provide a market model, but it supplies exactly the conceptual frame needed here: predictive success may generate distributional change rather than merely reveal it.

Fourth, empirical finance provides a practical motivation. McLean and Pontiff (2016) document that return predictors tend to weaken outside their discovery sample and weaken further after publication. Khandani and Lo (2011) document the August 2007 quant crisis as an extreme realisation of strategy crowding. MacKenzie and Millo (2003) and MacKenzie (2006) provide the canonical case study of a financial model (Black–Scholes) reshaping the very market it described. That evidence does not prove a single mechanism, but it is consistent with the possibility that awareness and imitation reduce predictability. The present proposal does not claim to identify those empirical effects directly. Instead, it builds a clean market laboratory for one plausible mechanism behind them.

2.3 Positioning and contribution

The gap in the literature is not the absence of intuition. Economists, finance researchers, and machine-learning theorists already recognise that behaviour can change predictive environments. The unresolved issue is operational: there is still limited evidence from a transparent and disciplined market simulation showing how a shared forecasting rule can diffuse, alter market dynamics, and lose value *because of* that diffusion.

This proposal contributes by narrowing the problem to its most interpretable form. It studies one risky asset, one public forecasting rule, and a small number of adoption mechanisms. The narrowing is not a loss of ambition; it is what makes the mechanism identifiable and the results interpretable for an interdisciplinary review panel. The proposal is therefore positioned between very abstract theory and highly elaborate market simulation. Its aim is to deliver a tractable framework in which reflexive forecast adoption can be examined directly.

3. Research design and methodology

3.1 Baseline market environment

The baseline combines two well-established building blocks from the heterogeneous-agent literature. Individual trader demand is microfounded as in Brock and Hommes (1998): each trader is a myopic mean-variance maximiser with constant absolute risk aversion (CARA). Price formation follows the disequilibrium market-adjustment tradition of Beja and Goldman (1980), formalised in discrete log-linear form by Farmer and Joshi (2002): a risk-neutral market maker absorbs net order flow and shifts the log-price proportionally to it.

Using these two ingredients places the proposal inside an established tradition while keeping the baseline minimal. Richer ingredients of the Brock–Hommes programme, in particular heterogeneity across trader types and multi-rule discrete-choice switching, are deferred to §3.7 so that the baseline remains a clean laboratory.

Asset and agents

Time is discrete and indexed by $t = 0, 1, \dots, T$. The market contains one risky asset with log-price p_t and one-period return $r_{t+1} = p_{t+1} - p_t$. The risk-free rate is normalised to zero ($R = 1$) and the asset pays no dividend, so r_{t+1} is interpreted as an excess return. This is the zero-dividend special case of Brock and Hommes (1998, Eqs. 2.1–2.3) and is the same accounting convention used by Farmer and Joshi (2002). There are N traders indexed by $i = 1, \dots, N$.

Trader demand (Brock–Hommes microfoundation)

Following Brock and Hommes (1998, Eqs. 2.1–2.3), each trader has CARA utility and maximises a one-period mean-variance objective over end-of-period wealth. With conditional forecast $\hat{r}_{i,t+1}$ of next-period return and common perceived conditional variance $\sigma^2 > 0$, the optimal signed position is

$$z_{i,t} = \frac{\hat{r}_{i,t+1}}{a \sigma^2}, \quad (1)$$

where $a > 0$ is the coefficient of absolute risk aversion. Equation (1) is the zero-dividend, zero-rate specialisation of the BH98 demand function $z_{ht} = E_{ht}[p_{t+1} + y_{t+1} - R p_t] / (a \sigma^2)$.

Economic meaning. A trader buys when expected next-period return is positive and sells when it is negative. Position size scales inversely to risk aversion and perceived variance, so the same forecast generates smaller trades when the market is perceived as riskier.

Adoption indicator and trader order

Let $a_{i,t} \in \{0, 1\}$ indicate whether trader i has adopted the advanced forecasting rule at time t . The trader's final order combines the null rule (§3.3) and the advanced rule (§3.4):

$$q_{i,t} = (1 - a_{i,t}) q_{i,t}^{(0)} + a_{i,t} q_{i,t}^{(A)}, \quad (2)$$

with the advanced order

$$q_{i,t}^{(A)} = \text{clip}\left(\frac{\hat{r}_{t+1}^{(A)}}{a \sigma^2}, -\bar{q}_i, \bar{q}_i\right). \quad (3)$$

Economic meaning. Non-adopters trade according to the benchmark rule. Adopters follow the microfounded mean-variance demand (1) evaluated at the public forecast $\hat{r}_{t+1}^{(A)}$ rather than at an idiosyncratic forecast. The position cap $\bar{q}_i > 0$ plays the role of a leverage or inventory constraint: it prevents unboundedly large trades when rolling-window estimates of $\hat{r}^{(A)}$ are noisy.

Price formation (market maker)

Price is set by a risk-neutral market maker who absorbs net order flow and shifts the log-price proportionally, as in Beja and Goldman (1980, Eqs. 2.1 and 2.6) and Farmer and Joshi (2002, Eq. 7). Aggregate signed demand is

$$D_t = (1/N) \sum_{i=1}^N q_{i,t}, \quad (4)$$

and the market maker updates the log-price according to

$$p_{t+1} = p_t + \mu D_t + \sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, 1), \quad (5)$$

where $\mu = 1/\lambda > 0$ is the market-maker price-impact coefficient and $\sigma > 0$ controls exogenous news. Equation (5) is the zero-fundamental specialisation of Farmer and Joshi (2002, Eq. 7).

Economic meaning. When aggregate signed demand is positive, the market maker absorbs the imbalance and raises the quote; when it is negative, the quote is lowered. The parameter μ measures the *linear price impact* of aggregate demand and is explicitly derivable from dealer inventory dynamics and from the path-independence axiom in Farmer and Joshi (2002, §2.2).

Baseline return equation

Taking first differences of (5) gives the one-period return process

$$r_{t+1} = \varphi r_t + \mu D_t + \sigma \varepsilon_{t+1}, \quad (6)$$

with $|\varphi| < 1$ small. The φr_t term is not ad hoc: Farmer and Joshi (2002, Eqs. 8–11) show analytically that when a fraction of aggregate demand comes from linear trend-following strategies, the induced return process has first-lag autocorrelation $\rho_r(1) = \alpha/(1 + \alpha) > 0$, of the same order as trend-following intensity α . Equation (6) absorbs a small, empirically plausible trend-following component in background order flow as a reduced-form AR(1) coefficient rather than modelling it as a separate trader type at baseline.

Economic meaning. The component φr_t introduces weak short-run predictability that an adaptive forecasting rule can, in principle, exploit. The component μD_t is the channel through which adoption feeds back into prices: when many adopters trade coherently on the shared forecast, aggregate demand concentrates in the same direction, the market maker absorbs it, and part of the predicted movement is realised within the current period. The predictable component available on the *next* period is therefore reduced. Through this channel, the diffusion of a shared rule can erode its own predictive content.

Profit

For evaluation, one-period profit of trader i is

$$\Pi_{i,t+1} = q_{i,t} r_{t+1}. \quad (7)$$

Transaction costs are omitted in the baseline and added as an extension in §3.7 so that forecast erosion can first be isolated as a purely dynamical phenomenon.

3.2 A three-layer model of trader behaviour

It is useful to make the structure of trader behaviour explicit, because a common ambiguity in proposals of this kind is whether the "forecasting rule" is the whole agent or only one module of a larger decision. Here behaviour is decomposed into three layers:

1. **Belief layer.** What return does the agent expect? The advanced rule uses a rolling autoregressive forecast $\hat{r}_{t+1}^{(A)}$ (§3.4). The null rule has no belief about returns and generates order flow that is directionally uninformative on average (§3.3).
2. **Decision layer.** How does that belief become a position? For adopters, the belief is mapped to a position through Brock–Hommes mean-variance demand (1) and the position cap (3). For non-adopters, the null order enters directly.
3. **Switching layer.** How does an agent decide whether to use the shared rule? Two mechanisms are considered: stochastic diffusion (§3.5) and risk-adjusted performance-based switching (§3.5).

This decomposition keeps the forecasting rule separable from the decision and switching rules. It matters analytically because the mechanism of interest operates on the belief layer (public information) and transmits to returns through the decision and switching layers (private action). Separating the three makes it possible to say precisely where erosion is produced.

3.3 Null model of trading

The null rule is intentionally simple, because it serves as a reference point rather than as a realistic trading strategy. The proposed baseline is a no-skill random trading rule:

$$q_{i,t}^{(0)} = \xi_{i,t}, \quad \xi_{i,t} \sim \mathcal{N}(0, \sigma_q^2), \quad (8)$$

with independent draws across agents and time.

Economic meaning. In the absence of the advanced model, traders generate order flow that is directionally uninformative on average. This provides a transparent benchmark: any systematic predictability or profitability generated by the advanced rule can be assessed relative to a market in which trading is active but not forecast-driven.

This benchmark is useful for three reasons. First, it avoids attributing predictive content to an already sophisticated baseline. Second, it makes the emergence of coordinated behaviour easy to interpret. Third, it allows the advanced rule to be evaluated against a deliberately low-information reference point. A simple heuristic alternative, such as $q_{i,t}^{(0)} = \psi_i r_t$ with small random coefficients ψ_i , is retained as a sensitivity analysis, but the random rule is the cleanest starting point. A richer ecology of trend followers, contrarians, and fundamentalists (Farmer and Joshi 2002) is introduced as a second-stage extension (§3.7).

3.4 Advanced forecasting rule

The advanced rule is based on an autoregressive forecast estimated on a rolling window of recent returns. Let w denote the rolling-window length and p the autoregressive order. At time t , the public model estimates

$$r_{s+1} = c + \sum_{j=1}^p \phi_j r_{s+1-j} + u_{s+1}, \quad (9)$$

using only the observations in the window $s = t - w, \dots, t - 1$. This yields the out-of-sample forecast

$$\hat{r}_{t+1}^{(A)}(w, p) = \hat{c}_{t,w,p} + \sum_{j=1}^p \hat{\phi}_{j,t,w,p} r_{t+1-j}. \quad (10)$$

Economic meaning. The model estimates how recent returns help predict the next return, using only information that would have been available at time t . Rolling estimation allows the coefficients to adapt gradually if the return process changes.

The core implementation begins with AR(1) models and a small set of rolling windows, for example $w \in \{50, 100, 250\}$. That design is sufficient to compare short-memory and longer-memory estimation while keeping the model class highly interpretable. Higher-order AR models can be explored once the baseline is working, but they are not essential to the central mechanism. Penalized regressions, tree-based ensembles, and recurrent architectures are intentionally left outside the core design (Mullainathan and Spiess 2017). The purpose of the baseline is not to maximise predictive accuracy at all costs, but to establish whether reflexive adoption can be observed in the simplest forecasting environment that should, in principle, work.

3.5 Adoption and diffusion mechanism

Two mechanisms are considered by which traders may adopt the advanced rule. Both act on the switching layer of §3.2.

Stochastic diffusion

Non-adopters discover or adopt the public model with a fixed probability $\pi \in (0, 1)$ each period:

$$P(a_{i,t+1} = 1 \mid a_{i,t} = 0) = \pi, \quad (11)$$

with an optional small exit probability δ :

$$P(a_{i,t+1} = 0 \mid a_{i,t} = 1) = \delta. \quad (12)$$

Economic meaning. Adoption spreads gradually without explicitly modelling social learning or profitability comparisons. This mechanism is analytically and computationally simple, making it ideal for early debugging and for isolating the pure effect of wider use of a common signal.

Risk-adjusted performance-based diffusion

In the economically meaningful version, adoption depends on recent relative *risk-adjusted* success. Because the decision layer is mean-variance based, the switching score is constructed to be consistent with that preference structure. Let $CE(\cdot)$ denote the one-period certainty-equivalent return of a strategy,

$$CE_t^{(k)} = \bar{\Pi}_t^{(k)} - (a/2) \hat{\sigma}_t^{2,(k)}, \quad (13)$$

where $\bar{\Pi}_t^{(k)}$ and $\hat{\sigma}_t^{2,(k)}$ are the rolling mean and variance of profit for rule $k \in \{(o), (A)\}$ computed over a window of length W_A . Define the switching score

$$S_t = CE_t^{(A)} - CE_t^{(0)}. \quad (14)$$

The adoption probability is then

$$P(a_{i,t+1} = 1 \mid a_{i,t} = 0) = \Lambda(\alpha + \beta S_t), \quad (15)$$

where $\Lambda(z) = 1/(1 + e^{-z})$ is the logistic function, α is a baseline adoption term, and $\beta > 0$ controls sensitivity to recent performance.

Economic meaning. Agents size positions as if they care about risk, so they should also *adopt* strategies as if they care about risk. Using a certainty-equivalent score, rather than raw average profit, removes an internal inconsistency present in earlier formulations: a high-variance rule cannot appear attractive simply because it takes more risk. The specification (15) is the binary BH98 discrete-choice mechanism applied at the switching layer.

The stochastic mechanism is implemented first because it is easier to interpret, and is especially useful for identifying whether wider use of the forecast is *by itself* sufficient to reduce predictability. The performance-based mechanism is then added to study feedback between early success and later crowding.

3.6 Intra-period timing

The positive-sign market-impact term μD_t in (6) creates a subtle timing question: if adopter demand today pushes the quote up today, adoption can appear self-fulfilling rather than self-eroding unless the order of events within a period is made explicit. The proposal resolves this by fixing the following sequence inside each period:

1. Agents observe returns up to and including period t and form forecasts $\hat{r}_{t+1}^{(A)}$.
2. Adopters and non-adopters submit orders $q_{i,t}$ per (2)–(3).
3. The market maker absorbs aggregate demand D_t and moves the quote immediately, as in (5).
4. The exogenous news shock $\sigma \varepsilon_{t+1}$ and the residual autoregressive term φr_t realise.

Forecast success is evaluated against the *residual* return remaining after contemporaneous crowding has already moved price. This ensures that erosion arises through the standard economic channel: adopters do not observe a forecast and then cause the forecast to come true at their own trading price. They trade on it, move the current quote, and then evaluate their forecast against what is *left* of the predictable return.

3.7 Extensions and robustness

The proposal is deliberately structured around a core design and a set of optional extensions. The core consists of one asset, the return law in (6), the random null rule, a shared rolling AR(1) forecast, and one adoption mechanism. These elements are sufficient to test the main hypothesis.

Once the core is functioning, the following extensions are natural, ordered by priority for a tractable capstone:

- **Heterogeneous trader ecology (first extension).** Replace the purely random null rule with a small ecology of trend followers, contrarians, and fundamentalists,

as in Farmer and Joshi (2002). Trend followers generate short-run positive autocorrelation and amplify noise; value investors generate negative autocorrelation; mixed populations can offset linear autocorrelation while still producing excess volatility and clustered volatility. This provides a more believable background market into which the AR signal diffuses.

- **Transaction costs.** Subtract proportional trading costs from profits to evaluate whether modest statistical predictability survives economically.
- **Retraining frequency.** Compare slow versus frequent model updates to examine whether retraining delays, freezes, or accelerates the erosion mechanism (Mendler-Dünner et al. 2020; Kabra and Patel 2024).
- **Delayed information diffusion.** Allow only a fraction of agents to observe the forecast immediately, with delayed adoption by others.
- **Multiple forecasting models.** Allow agents to choose among several AR specifications or between econometric and machine-learning rules. A diversity-vs-monoculture comparison becomes possible.
- **Endogenous volatility.** Replace constant σ with a volatility process that rises with large aggregate demand or large recent returns.
- **Adoption frictions.** Introduce fixed costs, minimum trial periods, or heterogeneous willingness to switch rules.
- **Multiple assets.** Extend the framework to a small portfolio environment.

These extensions are valuable but are not required to answer the central question. They are treated explicitly as second-stage exercises so that the scope remains disciplined.

4. Metrics and expected results

The metric set is organised into three primary endpoints, a diagnostic set, and a threshold metric. This structure follows directly from the causal claim: adoption of the shared rule should degrade its statistical usefulness, its economic usefulness, and it should do so by producing measurable crowding.

4.1 Primary metrics

Statistical endpoint — forecast erosion

The headline statistical measure is out-of-sample forecast erosion. Three complementary versions are computed:

- **Rolling mean-squared forecast error**, reported relative to a low-adoption benchmark MSFE: $R_{\text{MSFE}}(t) = \text{MSFE}_t / \text{MSFE}_{\text{benchmark}}$.
- **Rolling out-of-sample R^2** , measured against a naive constant-mean forecast: $R^2_{\text{OOS}}(t) = 1 - \sum (r_{s+1} - \hat{r}_{s+1})^2 / \sum (r_{s+1} - \bar{r})^2$.
- **Rolling effective AR coefficient.** Estimate $\phi^{\text{eff}}(t)$ directly from the realised return series in a trailing window and test whether it falls as the adoption share A_t rises. This goes directly at the theoretical mechanism in (6): contemporaneous

absorption of the predictable component through μD_t must reduce the lagged coefficient carried into next-period returns.

Economic endpoint — net trading value

The headline economic measure is *net* profitability after transaction costs, plus a risk-adjusted measure. Two quantities are reported:

- **Net profit per period** of the advanced rule, reported relative to the null benchmark.
- **Risk-adjusted return.** Either the Sharpe ratio $\bar{\Pi}/\sigma_{\Pi}$, or, to remain consistent with the CARA decision layer, the certainty-equivalent return (13). Raw average profit is demoted to a diagnostic metric: a strategy can look profitable simply because it takes more risk, and that should not be mistaken for predictive value.

Crowding endpoint — position synchronization

The headline crowding measure combines an aggregate rate with a micro-level synchronization statistic:

- **Adoption share** $A_t = (1/N) \sum_i a_{i,t}$.
- **Position synchronization** across adopters. Two variants are reported: pairwise Pearson correlation of positions $\rho_q(t)$; and the fraction-on-same-side $F_{ss}(t) = \max(\text{share long}, \text{share short})$ among adopters. Synchronization is the cleanest observable measure of whether the signal has become crowded.

4.2 Diagnostic and market-dynamics metrics

The following are reported as diagnostics rather than as headline endpoints:

- Rolling return autocorrelation $\rho_r(1, t)$.
- Rolling return volatility and excess kurtosis.
- Turnover / aggregate trading volume.
- Sign accuracy of $\hat{r}^{(A)}$. This is intentionally *demoted* to a diagnostic, because in finance getting the sign right is not sufficient: return magnitudes may shrink while signs remain correct, and transaction costs can then erase the edge.

4.3 Threshold metric — critical adoption share A^*

A compact summary of the overall dynamic is the *critical adoption share* A^* , defined as the smallest adoption level at which at least one of the following is true:

- out-of-sample predictive power crosses zero ($R^2_{\text{OOS}} \leq 0$);
- net-of-cost profitability of the advanced rule crosses zero;
- realised volatility jumps above a pre-registered threshold.

Reporting A^* across parameter regimes would convert the qualitative claim ("enough adopters erode the signal") into a quantitative statement ("erosion sets in at adoption shares above A^* for these values of μ , φ , and w "). That is the strongest compact headline the project can deliver.

4.4 Expected results

The central mechanism would be supported if the simulation produces the following pattern. First, the advanced rule should outperform the null benchmark when adoption is initially low. Second, the rule should diffuse under either stochastic or risk-adjusted performance-based adoption. Third, as adoption rises, positions should become more synchronized and aggregate demand should become more signal-driven. Fourth, out-of-sample forecast accuracy and risk-adjusted profitability should decline relative to the low-adoption phase. Fifth, the decline should be accompanied by measurable changes in market behaviour, such as a lower effective ϕ^{eff} , higher turnover, or greater volatility.

A particularly informative result would be a threshold pattern. If predictability remains fairly stable at low adoption but declines sharply once adoption exceeds A^* , that would provide strong evidence that crowding, rather than ordinary noise, is responsible for erosion. Even in the minimal model the logic is clear: if adopters trade proportionally to the same forecast, aggregate demand becomes more strongly linked to lagged returns, and the effective autoregressive coefficient carried into next-period returns is reduced by the market-impact term.

4.5 Scope and limitations

The proposal also admits meaningful null results. Forecast erosion may fail to appear if price impact is too weak, if adoption remains sparse, or if the predictive relation is robust enough to survive imitation. Such findings would still be scientifically useful: they would identify the conditions under which deployment-induced erosion is limited rather than pervasive. A well-specified null result is therefore not a failure of the design; it is evidence about the boundary of the mechanism.

At the same time, the conclusions should remain carefully bounded. Positive findings would support the proposition that endogenous erosion is a plausible and measurable mechanism in a class of single-asset markets. They would *not* establish that all return predictors fail for performative reasons, that the same mechanism explains every empirical anomaly, or that autoregressive models are realistic descriptions of actual trading practice. The proposal is about mechanism identification in a disciplined market laboratory, not universal explanation.

The main limitations are equally clear. The price-impact specification is reduced form. The market contains one asset and only a limited set of behaviours. The return process is stylised rather than calibrated to a specific dataset. These limitations are acceptable because the proposal aims at tractability and interpretability first. They should, however, be stated explicitly so that the contribution is understood correctly: the project tests whether a focused reflexive mechanism can generate forecast erosion, not whether it reproduces the full complexity of real-world market microstructure.

5. Feasibility and implementation plan

The implementation strategy is staged so that each component can be verified before the next is added. This sequencing is central to feasibility.

Stage	Objective	Main output
1. Specification	Freeze equations (1)–(15), parameter ranges, and evaluation metrics; fix intra-period timing per §3.6.	Short design note specifying notation, parameters, and comparison regimes.
2. Baseline simulator	Implement the single-asset market with return law (6), random null rule (8), Brock–Hommes demand (1), and reproducible simulation routines.	Working simulator that produces prices, returns, and aggregate demand under the null model.
3. Benchmark validation	Verify the null market behaves as intended, descriptive statistics are stable across runs, and no adoption-free variant shows drift in φ^{eff} .	Validated benchmark environment.
4. AR forecasting rule	Add rolling AR estimation, forecast generation, and forecast-based trading with position caps for $w \in \{50, 100, 250\}$.	Out-of-sample forecast and profit series for several rolling windows.
5. Adoption mechanism	Introduce first the stochastic adoption process (§3.5) and then the risk-adjusted performance-based rule (§3.5).	Time series of adoption, position synchronization, and forecast performance under diffusion.
6. Experiments	Compare low-adoption, high-adoption, and no-adoption regimes; vary μ , φ , and window length; locate A^* .	Core figures and tables documenting when forecast erosion does or does not emerge.
7. Evaluation	Summarise results, interpret boundary cases, and separate core findings from optional extensions.	Final report, sensitivity analysis, and a clear statement of scope and limitations.

The most important feasibility principle is to preserve a strict order of construction. The market simulator should function before the forecasting rule is added. The forecasting rule should function before adoption is added. Extensions should be attempted only after the baseline produces stable and interpretable results. This prevents the project from becoming overly broad and ensures that any observed erosion can be traced to a small number of identifiable ingredients.

The empirical burden is also realistic. The baseline proposal relies on simulation rather than large-scale calibration or proprietary market data. Performance can be evaluated with standard and interpretable metrics: the primary endpoints in §4.1, together with the diagnostics in §4.2. These are sufficient for a first-pass assessment of the mechanism.

Finally, the proposal remains computationally manageable. Rolling autoregressive models are inexpensive to estimate, and the single-asset design keeps the state space small. That allows attention to remain on the scientific question rather than on engineering complexity.

6. Conclusion

This proposal advances a focused and feasible research programme on reflexive forecast adoption in financial markets. Its central claim is precise: when a forecasting rule becomes widely used, the resulting coordination of trading behaviour may alter the market dynamics that originally made the rule useful, thereby reducing its predictive and economic value. The project studies that possibility in the simplest setting where the mechanism can be examined transparently: a single-asset agent-based market with Brock–Hommes mean-variance demand, a Beja–Goldman / Farmer–Joshi market maker, a shared autoregressive forecasting rule, and explicit adoption dynamics.

The proposal is intentionally disciplined. It narrows a broad intellectual agenda into a design that is interpretable, implementable, and relevant to reviewers from multiple fields. Its originality lies not in asserting that markets are adaptive, but in operationalising a specific mechanism by which predictive success can generate its own erosion, and in measuring that mechanism through a tightened set of primary endpoints: out-of-sample forecast loss, net trading value, and position synchronization, with a critical adoption share A^* as a compact summary statistic. If successful, the project will provide a tractable framework for distinguishing deployment-induced deterioration from ordinary forecast instability, clarifying when model crowding matters, and identifying the conditions under which predictive systems remain robust or become self-undermining.

References

- Axtell, R. L. and J. D. Farmer (2025). "Agent-Based Modeling in Economics and Finance: Past, Present, and Future." *Journal of Economic Literature* 63(1), 197–287. DOI: 10.1257/jel.20221319.
- Beja, A. and M. B. Goldman (1980). "On the Dynamic Behavior of Prices in Disequilibrium." *Journal of Finance* 35(2), 235–248. DOI: 10.1111/j.1540-6261.1980.tb02151.x.
- Brock, W. A. and C. H. Hommes (1998). "Heterogeneous Beliefs and Routes to Chaos in a Simple Asset Pricing Model." *Journal of Economic Dynamics and Control* 22(8–9), 1235–1274. DOI: 10.1016/S0165-1889(98)00011-6.
- Brown, G., S. Hod, and I. Kalemaj (2022). "Performative Prediction in a Stateful World." In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics*, PMLR 151, 6045–6061.
- Farmer, J. D. and D. Foley (2009). "The Economy Needs Agent-Based Modelling." *Nature* 460, 685–686. DOI: 10.1038/460685a.
- Farmer, J. D. and S. Joshi (2002). "The Price Dynamics of Common Trading Strategies." *Journal of Economic Behavior & Organization* 49(2), 149–171. DOI: 10.1016/S0167-2681(02)00065-3.
- Giacomini, R. and B. Rossi (2009). "Detecting and Predicting Forecast Breakdowns." *Review of Economic Studies* 76(2), 669–705. DOI: 10.1111/j.1467-937X.2009.00545.x.
- Grazzini, J., M. G. Richiardi, and M. Tsionas (2017). "Bayesian Estimation of Agent-Based Models." *Journal of Economic Dynamics and Control* 77, 26–47. DOI: 10.1016/j.jedc.2017.01.014.

- Hardt, M. and C. Mendler-Dünnner (2025). "Performative Prediction: Past and Future." *Statistical Science* 40(3), 417–436. DOI: 10.1214/25-STS986.
- Kabra, A. and M. Patel (2024). "The Limitations of Model Retraining in the Face of Performativity." arXiv:2408.08499.
- Khandani, A. E. and A. W. Lo (2011). "What Happened to the Quants in August 2007? Evidence from Factors and Transactions Data." *Journal of Financial Markets* 14(1), 1–46. DOI: 10.1016/j.fin-mar.2010.07.005.
- Lo, A. W. (2004). "The Adaptive Markets Hypothesis: Market Efficiency from an Evolutionary Perspective." *Journal of Portfolio Management* 30(5), 15–29. DOI: 10.3905/jpm.2004.442611.
- Lucas, R. E., Jr. (1976). "Econometric Policy Evaluation: A Critique." *Carnegie-Rochester Conference Series on Public Policy* 1, 19–46. DOI: 10.1016/S0167-2231(76)80003-6.
- MacKenzie, D. (2006). *An Engine, Not a Camera: How Financial Models Shape Markets*. MIT Press. DOI: 10.7551/mitpress/9780262134606.001.0001.
- MacKenzie, D. and Y. Millo (2003). "Constructing a Market, Performing Theory: The Historical Sociology of a Financial Derivatives Exchange." *American Journal of Sociology* 109(1), 107–145. DOI: 10.1086/374404.
- McLean, R. D. and J. Pontiff (2016). "Does Academic Research Destroy Stock Return Predictability?" *Journal of Finance* 71(1), 5–32. DOI: 10.1111/jofi.12365.
- Mendler-Dünnner, C., J. Perdomo, T. Zrnic, and M. Hardt (2020). "Stochastic Optimization for Performative Prediction." In *Advances in Neural Information Processing Systems* 33.
- Mullainathan, S. and J. Spiess (2017). "Machine Learning: An Applied Econometric Approach." *Journal of Economic Perspectives* 31(2), 87–106. DOI: 10.1257/jep.31.2.87.
- Perdomo, J. C., T. Zrnic, C. Mendler-Dünnner, and M. Hardt (2020). "Performative Prediction." In *Proceedings of the 37th International Conference on Machine Learning*, PMLR 119, 7599–7609.
- Rossi, B. (2021). "Forecasting in the Presence of Instabilities: How We Know Whether Models Predict Well and How to Improve Them." *Journal of Economic Literature* 59(4), 1135–1190. DOI: 10.1257/jel.20201479.